



Instance segmentation and multi-object tracking for fruit quality grading and dynamic yield estimation in Egyptian citrus orchards

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ABSTRACT

Egypt ranks among the world's leading citrus producers and exporters; however, orchard operations remain constrained by subjective, labor-intensive manual quality grading and time-consuming yield estimation that require specialized expertise. This study presents an AI-driven orchard-intelligence framework for automated fruit quality grading and dynamic yield estimation under the unstructured conditions of Egyptian orange farms. To address the lack of in-field public datasets, a domain-specific instance-segmentation dataset of 618 original pixel-level annotated images, expanded to 1,545 augmented training images, and 15 videos was constructed. This dataset was used to benchmark single-stage segmentation models (YOLOv8, YOLOv11) against a two-stage (Mask R-CNN), alongside a transformer-based bounding-box detection (RT-DETR), under dense canopy occlusion, clustered fruit instances, and variable natural illumination. Each fine-tuned model was evaluated across two stages: static quality grading and dynamic yield estimation. For static grading (Good vs. Defective), YOLOv11 ranked among the top tier of evaluated segmenters, reaching mean Average Precision (mAP) of 0.932 with balanced precision-recall behavior, while maintaining real-time feasibility at 26.1 ms per image. For dynamic yield estimation, a multi-object tracking pipeline was implemented using BoT-SORT with camera-motion compensation and a virtual line-crossing counting mechanism. The evaluation reveals a three-way dissociation: the highest-mAP static segmenter (YOLOv11), the best dynamic total counter (RT-DETR, WMAPE = 5.66%, despite the lowest box-level static accuracy), and the best quality-resolved dynamic grader (YOLOv11s, WMAPE = 6.97% at 14.2 ms per frame) are three different systems; RT-DETR did not resolve fruit quality reliably under motion. Counting accuracy and grading capability are thus separable properties requiring separate evaluation. Overall, the framework enables objective fruit quality assessment and actionable in-field fruit counting to support harvest logistics and labor planning, providing a practical foundation for smart agriculture in the Egyptian citrus sector.

1. Introduction

Global food demand is projected to intensify as the world population approaches 10 billion by 2050 [1], requiring sustainable intensification that increases productivity without proportional resource consumption. Within this paradigm, citrus is a strategic crop: Egypt is a dominant orange exporter, with annual shipments of approximately 2 million metric tons [2] and a significant share of national agricultural exports [3]. As shown in Fig. 1, Egypt consistently leads the other major exporting origins in annual orange export volume, underscoring its strategic importance in the global citrus supply chain.

Despite this economic importance, Egyptian citrus production still depends heavily on manual fruit inspection, grading, and yield estimation. These practices are slow, subjective, and difficult to scale, and they are further strained by a steadily declining agricultural workforce

(Fig. 2) that raises production costs and operational uncertainty [4]. Manual yield estimation also introduces substantial errors that compromise harvest planning, labor allocation, and post-harvest logistics [5].

Artificial Intelligence (AI) and Computer Vision (CV) have emerged as key enablers of smart agriculture, automating fruit detection, localization, and quality assessment directly from images or videos and achieving substantial gains under occlusion, clustering, and variable illumination [6]. However, most existing work evaluates models on static images using segmentation accuracy metrics such as mean Average Precision (mAP), while practical yield estimation is performed on moving-camera video streams where temporal consistency, object tracking, and identity preservation become decisive. Consequently, high per-image segmentation accuracy does not automatically translate into reliable counting performance.

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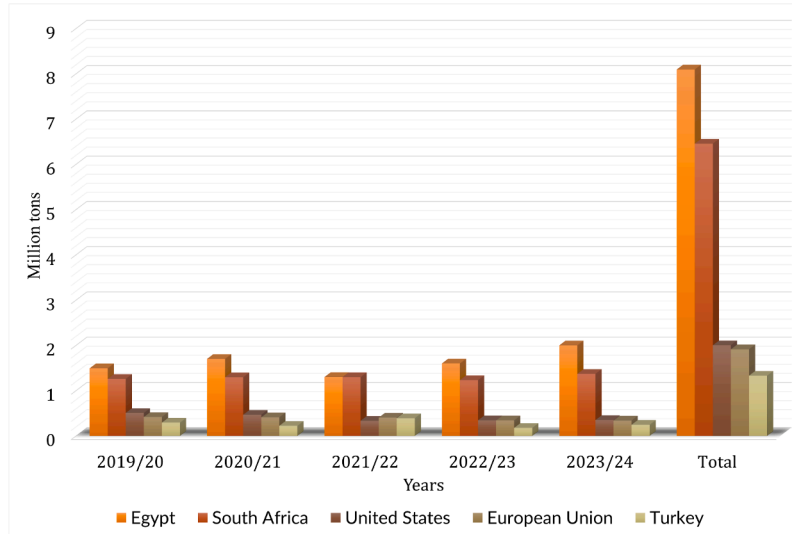


Fig. 1. Annual orange exports by the top 5 origins [2].

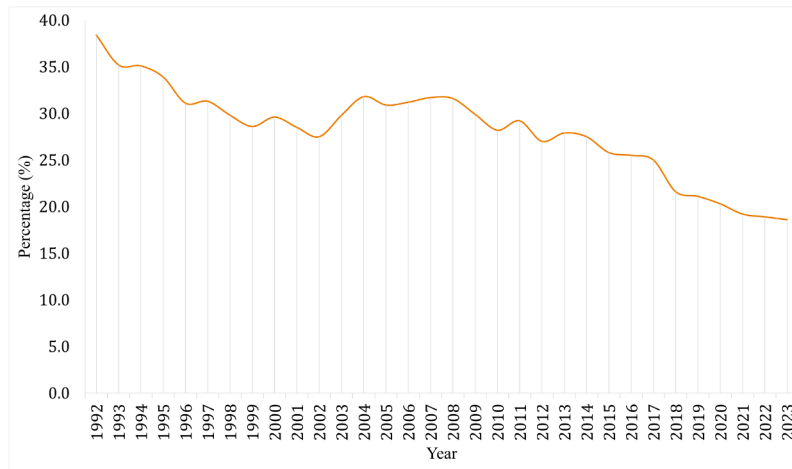


Fig. 2. Percentage of employment in the agricultural sector in Egypt [4].

1.1. Problem statement

Deploying computer vision in real Egyptian citrus orchards is challenging: fruits are frequently occluded by leaves and branches, clustered with partial overlap, and captured under variable natural illumination. Moreover, yield assessment is typically conducted using handheld moving-camera videos rather than framed still images, so intermittent detections and confidence oscillations can fragment tracked identities and cause systematic overcounting or undercounting. The core problem addressed in this study is therefore the lack of a field-validated, end-to-end pipeline that performs instance segmentation, quality grading, and dynamic yield estimation under the unstructured conditions of Egyptian citrus farms.

1.2. Research objectives

The primary objective is to develop and validate an end-to-end computer vision system for automated citrus crop monitoring and yield estimation. The specific objectives are:

- **Construct a domain-specific instance segmentation dataset** for Egyptian citrus, with 618 original images (expanded to 1,545 augmented training images) annotated into Good and Defective quality classes.

- **Benchmark YOLOv8, YOLOv11, and Mask R-CNN**, together with a transformer-based bounding-box detection baseline (RT-DETR), on the dataset using mAP@0.5, precision-recall behavior, and inference latency to determine the best accuracy-efficiency trade-off.
- **Validate dynamic yield estimation** by integrating the fine-tuned models with a multi-object tracker and evaluating counting reliability on moving-camera videos using Relative Error and Weighted Mean Absolute Percentage Error (WMAPE).

2. Related work

Deep learning methods for fruit recognition generally fall into two families. Single-stage detectors perform classification and localization jointly in one forward pass and favor inference speed; representative examples include YOLO [7], RetinaNet [8], and SSD [9]. Two-stage detectors first generate region proposals and then refine them, typically achieving higher accuracy at higher computational cost, with Fast R-CNN [10], Faster R-CNN [11], and Mask R-CNN [12] being the most widely adopted. More recently, real-time detection transformers such as RT-DETR [13] have eliminated non-maximum suppression through end-to-end set prediction; however, they operate at the bounding-box level and do not natively produce the instance masks required for pixel-level grading. Within orchard-oriented research, instance segmentation is increasingly preferred over bounding-box detection because its

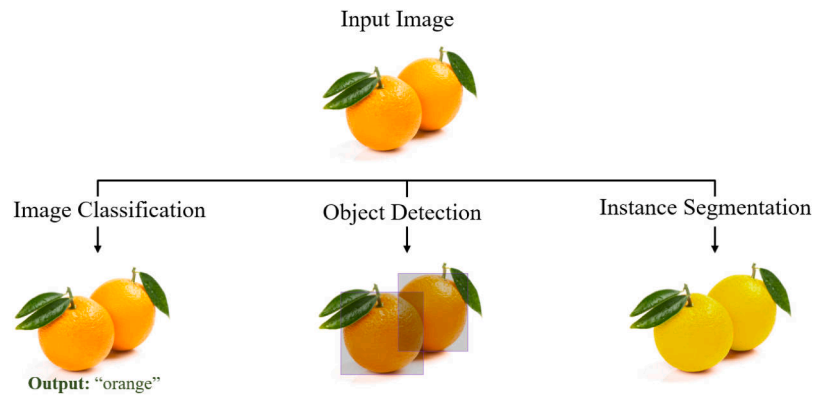


Fig. 3. Comparison between classification, object detection, and instance segmentation.

pixel-level masks enable reliable separation of clustered or occluded fruits and support shape-based analysis such as sizing [14].

2.1. From detection to instance segmentation

As illustrated in Fig. 3, image classification, object detection, and instance segmentation form a hierarchy of increasing spatial specificity. Bounding boxes provide rough localization but overlap heavily in clustered canopies, whereas instance masks separate adjacent fruits and capture shape information critical for accurate counting and sizing [14].

Early fruit detection work relied on YOLO variants, including YOLOv4 for orange detection [15] and several improved YOLO-based models for citrus [16–20], mangoes [21], apples [22], passion fruit [23], and capsicum [24]. More recently, YOLOv11 has demonstrated efficient real-time crop detection with clear gains over earlier versions [25]. However, bounding-box outputs remain insufficient whenever fruits overlap or when shape information is needed.

To address these limitations, instance segmentation has gained traction. Mask R-CNN has served as a benchmark for high-accuracy segmentation on sweet pepper [26], cocoa pods [27], strawberries [28], and partially occluded mangoes [29], but its two-stage design is computationally heavy for real-time edge deployment. Single-stage segmentation models such as YOLOv8-seg have extended YOLO with a mask branch, producing strong results on strawberries [30], tomatoes [31], weeds [32], and apple sizing via 3D reconstruction [33]. Notably, Sapkota et al. [6] showed that YOLOv8 outperformed Mask R-CNN on immature green apples (mAP@0.5 of 90.2% vs. 85.0%) while also being faster, suggesting that modern single-stage segmenters can offer a strong speed-accuracy trade-off for dense orchards.

2.2. Multi-class attribute assessment

Beyond localization, segmentation models have been extended to classify fruit attributes such as ripeness, growth stage, and quality. Examples include four cocoa-pod ripening stages with YOLOv8x [27]; ripe vs. unripe Gannan navel oranges with YOLOv5-NMM at 93.2% accuracy [20]; ripe vs. unripe strawberries with YOLOv8 [30]; five capsicum growth stages [24]; and healthy vs. diseased tomato plants [31]. Multi-class segmentation has also been used to isolate target fruits from surrounding plant structures relevant to robotic navigation or harvesting, such as apples vs. branches and trunks [6], sweet peppers and their peduncles [26], and uncut weeds vs. trunks and poles [32]. These studies support the feasibility of training a segmentation model that localizes citrus fruits and simultaneously classifies each instance as Good or Defective.

2.3. Yield estimation

Early yield estimation approaches relied on counting detections in static images [15,22,29,30]. Although useful for controlled evaluation, static-image counting fails in real orchards where moving cameras generate continuous video streams and the same fruit may be detected across many frames, causing severe overestimation [17]. Practical yield estimation therefore requires Multi-Object Tracking (MOT) to link detections of the same object over time. Representative tracking-based pipelines include centroid-based counting for mangoes [21], YOLOv5s + DeepSORT with virtual-line counting for citrus [17], tracking-based grape-cluster counting [34], and YOLOv8n + OC-SORT for passion fruit [23].

2.4. Summary of literature gaps

Table 1 summarizes recent computer vision studies in agriculture and highlights several important gaps. While foundational approaches such as object detection and instance segmentation are widely used, most studies adapt these methods to address only one agricultural objective, rather than developing integrated end-to-end pipelines. Detection-based approaches remain common because of their efficiency, but they do not provide the pixel-level precision required for handling clustered or occluded fruits. In addition, attribute-based multi-class classification, such as fruit quality grading, has received comparatively limited attention. Yield estimation is also still performed mainly on static images, which does not address repeated counting of the same fruit across consecutive video frames. Moreover, despite Egypt's importance as a citrus-producing and exporting country, only limited work has focused on the specific environmental and orchard conditions of Egyptian citrus farms. These gaps provide the main motivation for the end-to-end pipeline proposed in this study.

3. Methodology

The proposed framework is organized into three phases, as illustrated in Fig. 4: (i) construction of a domain-specific dataset, (ii) comparative training of single-stage and two-stage segmentation models, and (iii) implementation of a dynamic yield estimation pipeline. The following subsections detail each phase and the evaluation metrics used.

3.1. Dataset construction

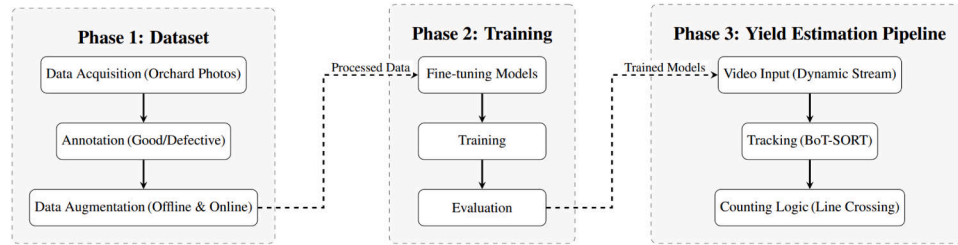
3.1.1. Data acquisition

Field data were collected from a commercial citrus farm in the Beheira Governorate, Egypt (30.5702° N, 30.2131° E), representative of the

Table 1

Summary of recent studies on computer vision tasks in agriculture, highlighting the proposed framework.

Task	Year	Ref	Crop	Dataset Size	Segmentation	Dynamic	Model	Number of Classes
Detection	2022	[16]	Orange	1266	×	×	Citrus-YOLOv7	1
Detection	2024	[18]	Orange	1223	×	×	YOLOv5	1
Detection	2024	[19]	Orange	975	×	×	AG-YOLOv8	1
Detection	2024	[20]	Orange	3000	×	×	YOLOv5-NMM	Ripe vs. Unripe
Detection	2025	[25]	Cabbage	2715	×	×	YOLOv11	1
Segmentation	2023	[26]	Pepper	1127	✓	×	Mask R-CNN	Fruit vs. Stem
Segmentation	2023	[31]	Tomato	1600	✓	×	YOLOv8	Healthy vs. Diseased
Segmentation	2024	[6]	Apple	1553	✓	×	YOLOv8 / Faster R-CNN	Fruit vs. Branch
Segmentation	2024	[33]	Apple	1079	✓	×	YOLOv8	1
Segmentation	2024	[32]	Weeds	5000	✓	×	YOLOv8	Weed vs. Tree
Segmentation	2025	[27]	Cocoa	4116	✓	×	YOLOv8 / Faster R-CNN	4 Ripening Stages
Segmentation	2025	[28]	Strawberry	3100	✓	×	Mask R-CNN	1
Yield Estimation	2021	[15]	Orange	1568	×	×	YOLOv4	1
Yield Estimation	2023	[22]	Apple	829	×	×	YOLOv7-tiny	1
Yield Estimation	2023	[17]	Orange	1940	×	✓	YOLOv5	1
Yield Estimation	2023	[34]	Grape	464	×	✓	YOLOv5	1
Yield Estimation	2024	[30]	Strawberry	813	✓	×	YOLOv8	Ripe vs. Unripe
Yield Estimation	2024	[29]	Mango	308	✓	×	Mask R-CNN	1
Yield Estimation	2024	[24]	Capsicum	450	✓	✓	YOLOv8	5 Growth Stages
Yield Estimation	2025	[23]	Passion Fruit	6445	×	✓	YOLOv8	1
Yield Estimation	2025	[21]	Mango	704	×	✓	YOLOv5	1
Yield Estimation (Proposed Framework)	2026	–	Orange	618 (1545 augmented training images)	✓	✓	YOLOv8 / YOLOv11 / Mask R-CNN / RT-DETR	Good/s. Defective

**Fig. 4.** Proposed research methodology framework.**Table 2**

Software environments used for reproducibility (YOLO vs. Mask R-CNN).

Component	YOLO Environment	Mask R-CNN Environment
Python	3.11.0	3.10.11
PyTorch	2.7.0	1.13.1
CUDA	12.6	11.6
cuDNN	9.7.1	8.3.2
Ultralytics	8.3.143	-
Detectron2	-	0.6

Egyptian citrus sector. Acquisition took place in the pre-harvest season when fruits had reached commercial maturity and full coloration. A total of 618 high-resolution images (3024×4032 pixels) were captured with a smartphone camera from multiple angles and distances. Close-up shots captured fine fruit details and surface defects; wider shots documented full-tree fruit distribution; oblique angles were used deliberately to include oranges partially hidden inside the canopy. Representative samples are shown in Fig. 5.

3.1.2. Data annotation

Images were manually annotated in Roboflow using polygonal masks to exclude background noise and preserve exact fruit contours. Each fruit instance was labeled as “Good” or “Defective.” A fruit was assigned the Defective class if it exhibited (i) visible disease symptoms, such as citrus canker lesions, black spot, or rind rot; (ii) physical damage, including punctures, splits, or crushing; or (iii) clearly abnormal discoloration or deformation; all remaining fruits were labeled Good. To control anno-

tation subjectivity, the entire dataset was labeled independently by two annotators, and every annotation was subsequently verified by a third reviewer who adjudicated all disagreements before the dataset was finalized. The dataset, training images, contains 376 Good and 239 Defective instances (615 in total, prior to any offline or online augmentation), corresponding to a class ratio of approximately 61:39. Annotations were exported in both YOLO format (for YOLOv8/YOLOv11 and RT-DETR) and COCO JSON format (for Mask R-CNN). Fig. 6 shows the polygonal ground truth corresponding to the samples in Fig. 5.

3.1.3. Data augmentation

To preclude any data leakage, the 83.3:8.1:8.6 train/validation/test split was performed before augmentation. The original dataset of 618 images was partitioned into exactly 515 training images, 50 validation images, and 53 blind-test images. All augmentation described below was applied *only* to the 515-image training partition, while the validation and test subsets consist solely of original, non-augmented images. *Offline augmentation* in Roboflow expanded the training partition by a $3\times$ factor, yielding exactly $515 \times 3 = 1,545$ augmented training images. Geometric transformations (random rotations of $\pm 15^\circ$, horizontal and vertical flips) handled camera-orientation variability, while photometric transformations (random brightness and exposure adjustments) helped the model separate true defects from lighting artifacts such as shadows or direct sunlight (Fig. 7). *Online augmentation* during training followed each architecture’s best practice: YOLO models used Mosaic augmentation to improve small-object detection, while Mask R-CNN used random flipping and multi-scale resizing; aggressive techniques such as Mosaic were avoided for Mask R-CNN to preserve the structural context required by its Region Proposal Network (RPN).



Fig. 5. Sample of collected images from the constructed dataset; sub-figures (a)–(i) illustrate the variability in the data collection process.

3.2. Implementation and training configuration

3.2.1. Experimental environment

All training and evaluation were performed on a single workstation with an Intel Core i9-14900HX CPU (2.2 GHz, 32 threads), 16 GB of RAM, and an NVIDIA GeForce RTX 4070 GPU (8 GB VRAM). The YOLOv8 and YOLOv11 networks were trained with the Ultralytics package, and Mask R-CNN with Detectron2; two isolated virtual environments managed dependency conflicts. Exact software versions are listed in Table 2. To ensure a controlled comparison across all model configurations, the random seed was fixed (seed = 0) for every training run, so that all models were exposed to identical data ordering, augmentation sampling, and initialization stochasticity.

3.2.2. Hyperparameter configuration

Hyperparameters were tuned to the characteristics of the Egyptian citrus dataset. Full configurations are given in Table 3 for YOLO, Table 4 for Mask R-CNN, and Table 5 for RT-DETR.

YOLO configuration

YOLOv8 and YOLOv11 were trained with AdamW (base learning rate 1.667×10^{-3}), a 3-epoch warmup (bias LR 0.1, momentum 0.9), and cosine annealing thereafter. A weight decay of 0.0005 was applied to mitigate overfitting. The composite loss combined Box Regression, Classification, and Distribution Focal Loss, with gains biased toward accurate localization in dense canopies. Training was capped at 300 epochs with early stopping (patience = 50).

Mask R-CNN configuration

Mask R-CNN (Detectron2) used a ResNet-50 + FPN backbone and SGD (momentum 0.9, base LR 0.001, batch size 4 due to memory). Training ran for 6,400 iterations with a linear warmup over the first 640 iterations followed by a multi-step decay (factor 0.1 at iterations 768 and 1024). The RPN was configured to sample 128 RoIs per image to improve detection under clutter. The final checkpoint was selected by the lowest validation loss.

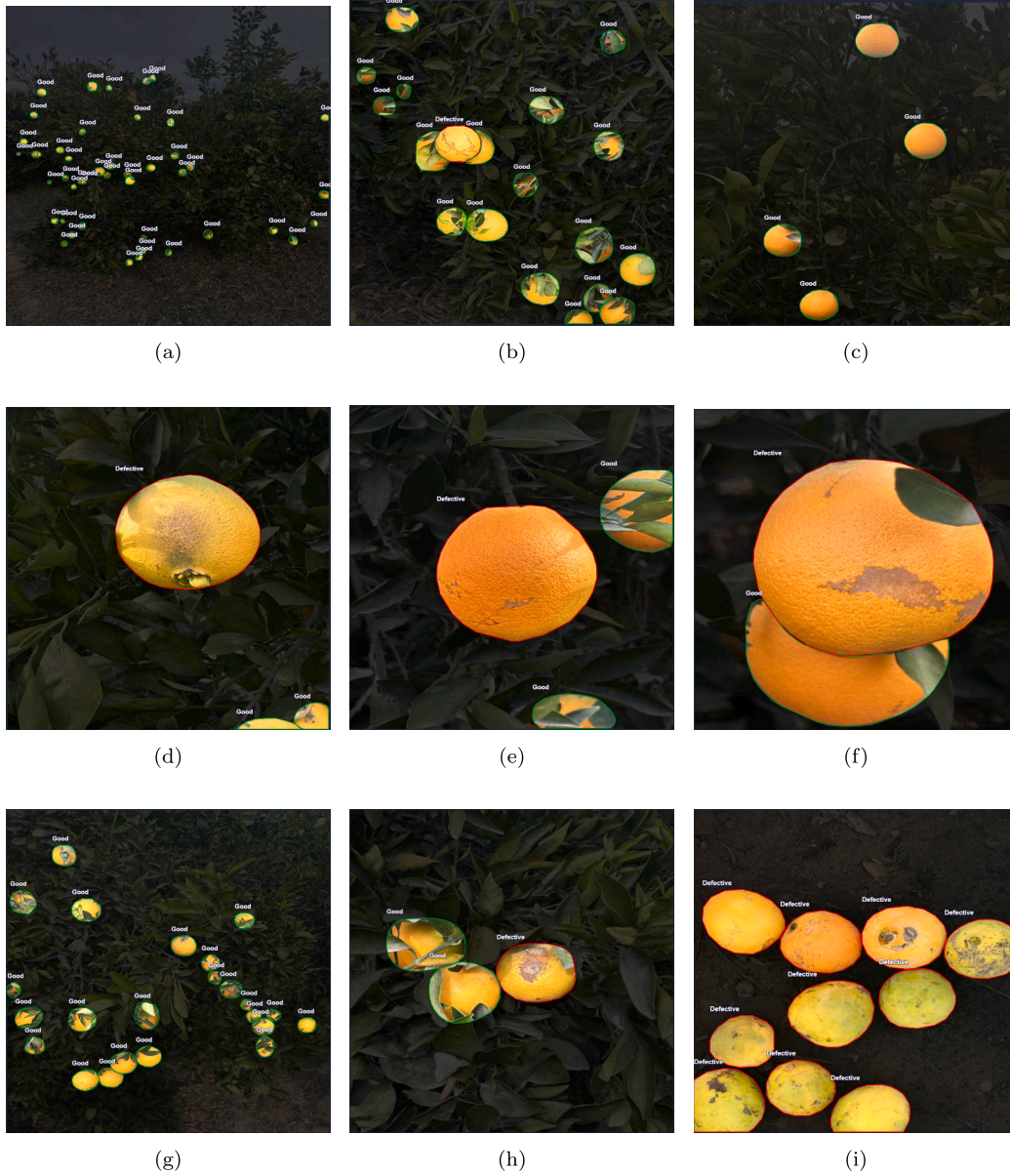


Fig. 6. Visualization of the annotated ground truth; sub-figures (a)–(i) display the polygonal masks overlaid on the original images, classifying oranges as either “Good” or “Defective”.

RT-DETR configuration

A transformer-based baseline, RT-DETR-1, was additionally fine-tuned on the dataset with the same fixed random seed (seed = 0), using the Ultralytics implementation within the same software environment as the YOLO models (Table 2). Training used AdamW and an early-stopping protocol, an input resolution of 640×640 , 300 object queries, and a GloU-based localization loss.

3.3. Dynamic yield estimation pipeline

Static image analysis is limited for real yield estimation because a moving camera revisits the same fruit, causing double counting, while single viewpoints often hide defects. The proposed dynamic pipeline (Fig. 8) extracts frames, applies instance segmentation (YOLO or Mask R-CNN), assigns persistent identities via multi-object tracking, and then counts each fruit exactly once as its trajectory crosses a virtual line. Every crossing also updates the corresponding Good or Defective counter,

yielding real-time quality-aware yield statistics. The tracking and counting components are detailed below.

3.3.1. Multi-object tracking integration

BoT-SORT was used to assign a unique persistent ID to each orange across frames. It predicts per-track positions with a Kalman filter and associates new detections by IoU overlap. Unlike simpler trackers that assume a static camera, BoT-SORT applies Global Motion Compensation (GMC) to counteract handheld camera motion, which is essential for field videos. The same tracker configuration (Table 6) was used for every segmentation model so that differences in counts reflect detector stability rather than tracker behavior.

3.3.2. Virtual line counting mechanism

For a frame width W , a vertical counting line is placed at:

$$x_{\text{line}} = 0.85 W \quad (1)$$

The value $0.85 W$ is an empirically selected parameter specific to the current validation protocol. It was chosen so that (i) the line lies

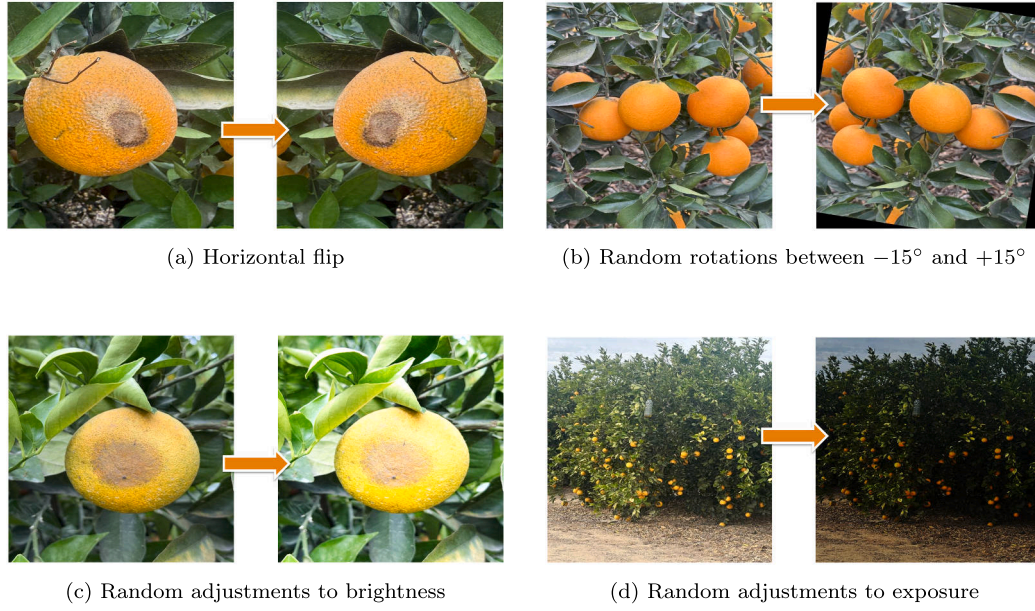


Fig. 7. Examples of offline data augmentation techniques applied to the dataset, including geometric transformations (a-b) and photometric transformations (c-d).

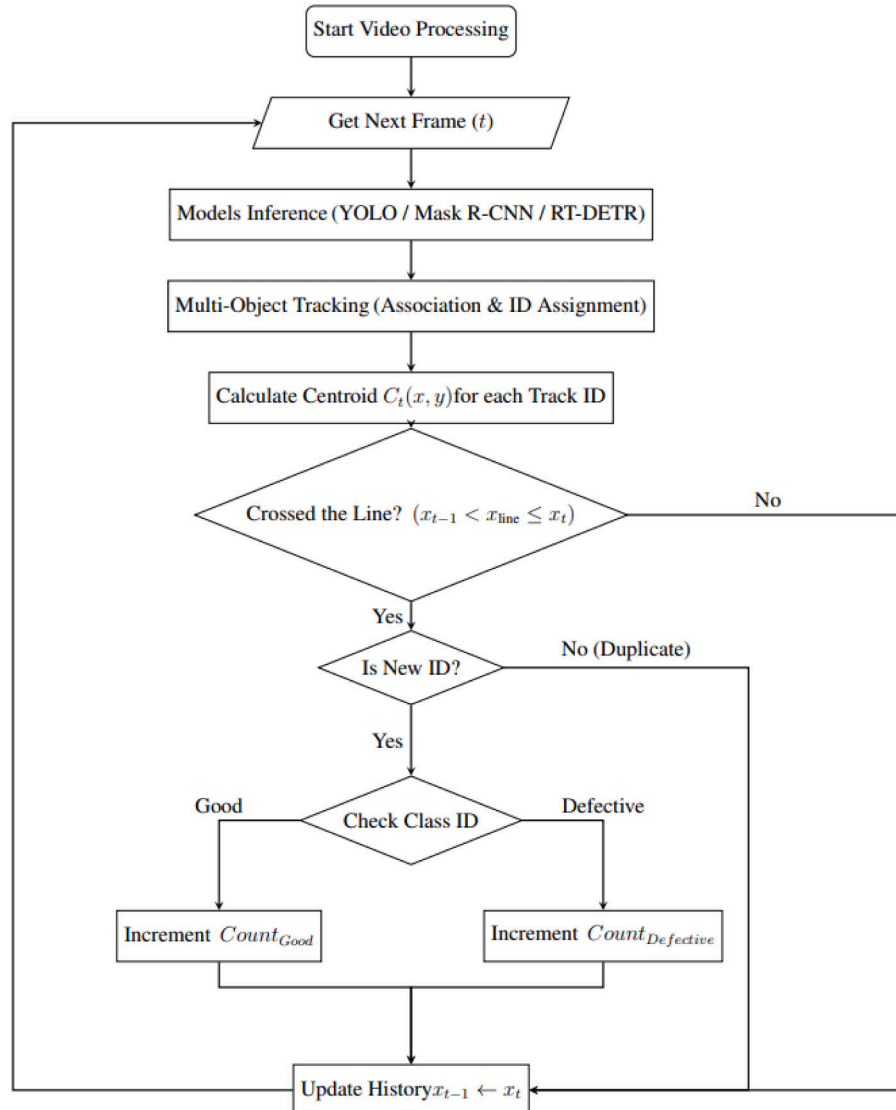


Fig. 8. Flowchart of the proposed dynamic yield estimation pipeline, illustrating the integration of segmentation, multi-object tracking, and counting logic.

Table 3

Training hyperparameters for the single-stage YOLO architectures.

Parameter	Value
Optimizer	AdamW
Initial LR (lr_0)	1.667×10^{-3}
Final LR Ratio (lrf)	0.01
Momentum	0.9
Weight Decay	0.0005
Max Epochs	300
Patience (Early Stopping)	50 Epochs
Batch Size	8
Warmup Epochs	3
Box Loss Gain	7.5
Class Loss Gain	0.5
DFL Loss Gain	1.5
IoU Threshold	0.7

Table 4

Training hyperparameters for the two-stage Mask R-CNN architecture.

Parameter	Value
Optimizer	SGD
Base LR	0.001
Momentum	0.9
Weight Decay	0.0001
Max Iterations	6400
Batch Size	4
Min Image Size	800
Warmup Iters	640
Gamma Decay	0.1
LR Decay Steps	[768, 1024]
RPN Batch/Img	128

Table 5

Training hyperparameters for the RT-DETR transformer architecture.

Parameter	Value
Optimizer	AdamW
Initial LR (lr_0)	1.667×10^{-3}
Final LR Ratio (lrf)	0.01
Momentum	0.9
Weight Decay	0.0005
Max Epochs	300
Patience (Early Stopping)	50 Epochs
Batch Size	4
Input Resolution	640×640
Warmup Epochs	3
Object Queries (max_det)	300
GlO U Loss Gain	7.5
Class Loss Gain	0.5

sufficiently far along the direction of camera travel for each track to have persisted for several frames before crossing, allowing BoT-SORT to consolidate the identity ($min_hits = 3$) and the detection confidence to stabilize, and (ii) it is not so close to the frame border that fruit centroids exit the frame before crossing it, given the lateral panning motion used in our acquisition protocol.

For each confirmed track id , the bounding-box centroid is tracked across frames.

$$x_t^{(id)} = \frac{x_{1,t}^{(id)} + x_{2,t}^{(id)}}{2}, \quad y_t^{(id)} = \frac{y_{1,t}^{(id)} + y_{2,t}^{(id)}}{2}. \quad (2)$$

A valid crossing is recorded when the centroid transitions from strictly left of the line in frame $t - 1$ to on or past the line in frame t :

$$\mathcal{E}_{id}(t) = (x_{t-1}^{(id)} < x_{line}) \wedge (x_t^{(id)} \geq x_{line}). \quad (3)$$

A binary flag $z_{id} \in \{0, 1\}$ enforces the count-once rule: it is set to 1 at the first crossing and prevents subsequent detections of the same id from incrementing the count. The total yield is

$$z_{id} = \begin{cases} 1, & \text{if the track crosses the counting line at any frame,} \\ 0, & \text{otherwise,} \end{cases}$$

$$\hat{N} = \sum_{id \in I} z_{id}, \quad (4)$$

where I is the set of all unique track identities. At each valid crossing, the fruit's class label routes the increment to either $Count_{Good}$ or $Count_{Defective}$.

3.4. Evaluation metrics

3.4.1. Segmentation performance

Segmentation accuracy was measured at an Intersection over Union (IoU) threshold of 0.5, where

$$IoU = \frac{|M_{pred} \cap M_{gt}|}{|M_{pred} \cup M_{gt}|} \quad (5)$$

Detections with $IoU \geq 0.5$ count as True Positives (TP); others are False Positives (FP), and missed fruits are False Negatives (FN). Precision, Recall, and F1-score are then defined [35] as:

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}, \quad F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}. \quad (6)$$

Across confidence thresholds, mean Average Precision at $IoU = 0.5$ aggregates per-class Average Precision over the two classes (Good and Defective):

$$mAP@0.5 = \frac{1}{C} \sum_{c=1}^C AP_c, \quad (7)$$

computed with the COCO evaluation protocol (Ultralytics for YOLO and Detectron2 COCOevaluator for Mask R-CNN). In the yield pipeline, high Precision limits false fruits added to the count and high Recall ensures occluded fruits are tracked through the scene; the latter is particularly important for the Defective class so damaged fruits are not misgraded.

3.4.2. Efficiency of models

Computational efficiency was quantified by the per-image total latency decomposed into three stages,

$$T_{total} = T_{preprocess} + T_{inference} + T_{postprocess}, \quad (8)$$

in addition to parameter count, network depth, and GFLOPs. Training practicality was characterized by total wall-clock time and number of epochs to convergence.

3.4.3. Yield estimation accuracy

The per-video estimation error captures both magnitude and direction (positive for over-counting, negative for under-counting):

$$Error(\%) = \left(\frac{C_{predicted} - C_{ground_truth}}{C_{ground_truth}} \right) \times 100. \quad (9)$$

Overall counting stability across videos is summarized with the Weighted Mean Absolute Percentage Error (WMAPE), which weights each video by its ground-truth count so that small-count videos do not dominate the result:

$$WMAPE = \frac{\sum_{i=1}^N |C_{predicted}^{(i)} - C_{ground_truth}^{(i)}|}{\sum_{i=1}^N C_{ground_truth}^{(i)}} \times 100, \quad (10)$$

where N is the number of evaluated videos.

Table 6
BoT-SORT configuration used for multi-object tracking in dynamic yield estimation.

Parameter	Value	Description
max_age	30	Maximum duration (frames) a track may remain lost before termination.
min_hits	3	Minimum matched frames required to initialize a confirmed track (suppresses short-lived false tracks).
iou_threshold	0.30	Minimum IoU overlap required for valid detection-to-track association.
gmc_method	sparseOptFlow	GMC using sparse optical flow features.
gmc_downscale	2	Downscale factor applied during GMC to improve efficiency while preserving stability.

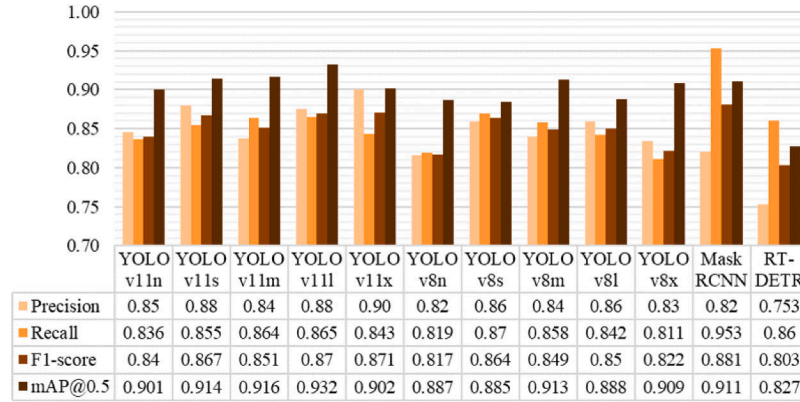


Fig. 9. Segmentation performance for all YOLOv8, YOLOv11, and Mask R-CNN configurations, together with the RT-DETR bounding-box baseline (box-level metrics).

4. Results and discussion

This section evaluates the trained models on the Egyptian citrus dataset, first on static segmentation and computational efficiency, and then on dynamic video-based yield estimation.

4.1. Segmentation performance

Fig. 9 summarizes the Precision, Recall, F1-score, and mAP@0.5 values of every evaluated model. All segmentation models exceed mAP@0.5 of 0.88, confirming that both single-stage and two-stage architectures can handle the occlusion and clustering typical of Egyptian orchards. YOLOv11l (large) was among the top tier of evaluated segmenters, attaining mAP@0.5 of 0.932 with balanced precision-recall behavior (Precision = 0.88, Recall = 0.865, F1 = 0.87). The transformer-based RT-DETR baseline recorded a Precision of 0.753, a Recall of 0.860, an F1-score of 0.803, and an mAP@0.5 of 0.827. Because RT-DETR functions as a bounding-box detector rather than a mask-level instance segmenter, these metrics are computed exclusively at the bounding-box level to serve as a cross-paradigm detection reference. Furthermore, its detection profile is notably recall-leaning (Recall 0.860 vs. Precision 0.753), a characteristic whose practical consequences for temporal tracking will be discussed in the dynamic evaluation of Section 4.6.

The YOLOv11 gain over YOLOv8 is attributable to architectural refinements, in particular the C3k2 bottleneck and C2PSA spatial-attention modules, which strengthen multi-scale feature processing in natural canopies with varied fruit sizes and filter out leaf/branch clutter, improving Precision.

4.1.1. Precision-recall trade-off

Mask R-CNN produced a notably different profile: a top-ranked Recall of 0.953 paired with a lower Precision of 0.82. This is characteristic of two-stage detectors, whose region-proposal stage produces many candidates, beneficial for Recall but more prone to false positives on leaf and background clutter. YOLO models apply stricter selection and therefore achieve higher Precision. Our findings extend those of Sapkota et al. [6] (YOLOv8 > Mask R-CNN on apples, 90.2% vs. 85% mAP): while

YOLOv11l ranked among the top tier on mAP, Mask R-CNN remains practical when maximizing Recall is the priority over strict quality grading.

4.1.2. Model complexity and performance scaling

Within the YOLOv11 family, accuracy improves up to the Large variant (mAP@0.5 = 0.932) and then drops for the Extra-Large variant (0.902). With 1,545 augmented training images, the extra capacity of YOLOv11x (62.0 M parameters vs. 27.6 M for the Large) is underutilized given the size of the training set; the observed drop is therefore best interpreted as a data-limited over-parameterization effect, whereby the larger model becomes more sensitive to training noise rather than learning better features, and its greater depth likely complicates optimization.

4.1.3. Class-wise quality grading analysis

To assess the suitability of the evaluated models for automated quality grading, segmentation performance was analyzed separately for the Good and Defective classes. The detailed class-wise results for all YOLO configurations are presented in Table 7.

A clear and consistent class-wise asymmetry was observed across nearly all evaluated models: defective fruits were detected more reliably than good fruits. The average Recall for the Defective class was approximately 0.93, whereas the average Recall for the Good class was substantially lower at 0.761. From a quality-control standpoint, this asymmetry is favorable, since it minimizes the risk of a defective fruit being missed and erroneously accepted as good. Conversely, the Good class exhibited a higher average Precision (0.905) than the Defective class (0.804), indicating that fruits classified as good were rarely mislabeled.

Comparing the two architecture families, YOLOv11 generally achieved stronger defect-detection performance than YOLOv8 at matched scales, with the exception of the extra-large variant. For instance, YOLOv11n attained a Defective mAP@0.5 of 0.933, exceeding YOLOv8n at 0.913, which suggests that the architectural refinements of YOLOv11 improve the learning of the fine visual patterns associated with fruit defects even at small model scales. The advantage widens at larger scales: YOLOv11l achieved a high Defective mAP@0.5 (0.952),

Table 7

Class-wise performance comparison (Good vs. Defective oranges) across YOLO architectures. The F1-score highlights the trade-off between Precision and Recall for each class.

Model	Class: Good				Class: Defective			
	Precision	Recall	F1-Score	mAP@0.5	Precision	Recall	F1-Score	mAP@0.5
YOLOv11n	0.838	0.797	0.817	0.868	0.853	0.875	0.864	0.933
YOLOv11s	0.936	0.734	0.823	0.903	0.823	0.975	0.893	0.926
YOLOv11m	0.842	0.844	0.843	0.903	0.835	0.884	0.859	0.928
YOLOv11l	0.943	0.781	0.854	0.912	0.806	0.950	0.872	0.952
YOLOv11x	0.988	0.703	0.821	0.891	0.814	0.982	0.890	0.913
YOLOv8n	0.881	0.688	0.773	0.860	0.751	0.950	0.839	0.913
YOLOv8s	0.920	0.766	0.836	0.879	0.797	0.975	0.877	0.891
YOLOv8m	0.894	0.790	0.839	0.904	0.786	0.925	0.850	0.922
YOLOv8l	0.929	0.734	0.820	0.868	0.789	0.950	0.862	0.908
YOLOv8x	0.876	0.772	0.821	0.893	0.791	0.850	0.819	0.926

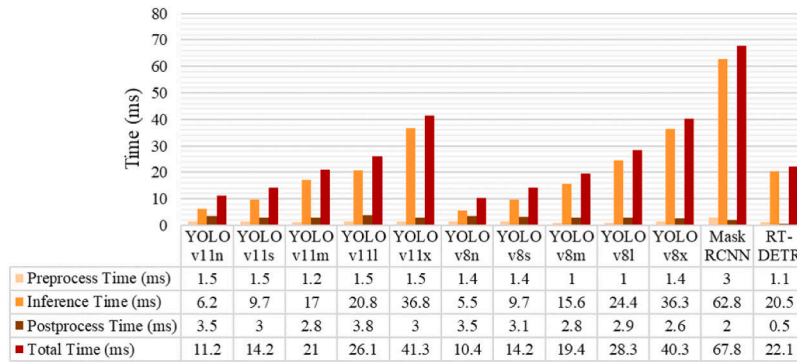


Fig. 10. Comparative analysis of preprocessing, inference, postprocessing, and total latency for all evaluated models, including the RT-DETR bounding-box baseline.

exceeding YOLOv8l (0.908). These results indicate that the YOLOv11 family is better suited to quality-grading tasks on this dataset.

The class-wise results also expose distinct operating trade-offs across model scales. YOLOv11s achieved the highest Defective Recall (0.975), making it the preferred choice when detecting every possible defect is the overriding priority, even at the cost of rejecting some good fruits. YOLOv11m offered the most balanced behavior between the two classes, making it a strong candidate for general-purpose grading. Consistent with the aggregate results in Section 4.1.2, YOLOv11x provided no improvement over YOLOv11l: despite a very high Precision on the Good class, its Defective-class performance regressed relative to the Large variant, reinforcing the conclusion that increasing capacity beyond the Large scale yields no benefit on a dataset of this size. Importantly, the uniformly high Defective Recall observed here (0.85–0.98 across all models) indicates that the static quality-grading capability of the trained models is consistent across architectures and scales; as discussed in Section 4.6, the low absolute Defective counts observed in the dynamic evaluation therefore reflect the low prevalence of defective fruits in harvest-season videos rather than a limitation of the grading models themselves.

4.2. Inference speed and real-time feasibility

Per-image latency, decomposed into preprocessing, inference, and postprocessing on an RTX 4070 (8 GB), is shown in Fig. 10, and the accuracy-latency trade-off is plotted in Fig. 11.

Mask R-CNN reached 67.8 ms (~14.7 FPS), dominated by its 62.8 ms inference stage, expected for a two-stage pipeline that proposes then refines regions. YOLOv11l processed the same image in 26.1 ms (~38.3 FPS), well above the ~30 FPS threshold used for real-time applications. The RT-DETR baseline required 22.1 ms per image (1.1 ms preprocessing, 20.5 ms inference, and 0.5 ms postprocessing; ~45 FPS).

In the accuracy-latency plot, models closer to the upper-left corner are preferred. YOLOv11l sits in this region, offering the most effective

balance: YOLOv11n and YOLOv11s are faster but less accurate, while YOLOv11x and Mask R-CNN are slower without a comparable accuracy gain. Within YOLOv11, latency grows from 11.2 ms (n) to 41.3 ms (x), confirming the expected capacity-speed trade-off but also showing that the largest variant is not the optimal deployment choice.

4.3. Architectural complexity and resource efficiency

Parameter count (Fig. 12), network depth (Fig. 13), and GFLOPs (Fig. 14) characterize the suitability of each model for resource-constrained platforms.

YOLOv11 is markedly more parameter-efficient than YOLOv8 at comparable scales. YOLOv11l uses only 27.6 M parameters, versus 45.9 M for YOLOv8l and 43.9 M for Mask R-CNN. For deployment this translates to a much smaller on-disk footprint (~55 MB for YOLOv11l vs. ~90 MB for YOLOv8l), easing integration into embedded systems. Compute cost follows the same trend: YOLOv11l requires 141.9 GFLOPs, roughly 46% below Mask R-CNN's 265.5 GFLOPs and well below YOLOv8l's 220.1 GFLOPs, important for battery-powered platforms. Although YOLOv11 is deeper than YOLOv8 (e.g., 203 vs. 125 layers at the Large scale), its design trades width for a lighter, more expressive structure. The RT-DETR occupies an intermediate complexity point: approximately 32.0 M parameters (between YOLOv11l's 27.6 M and Mask R-CNN's 43.9 M), 302 layers (the deepest network evaluated), and 103.4 GFLOPs (below YOLOv11l's 141.9 GFLOPs).

4.4. Training efficiency and convergence analysis

Training time and epochs to convergence are summarized in Fig. 15. Epoch count and wall-clock time do not strictly track one another: for instance, YOLOv11s ran for 188 epochs but finished in under 30 minutes, while the larger variants required substantially more time per epoch.

Convergence was not monotonic across model scales. YOLOv8s and YOLOv8m converged faster (94 and 88 epochs, respectively) than

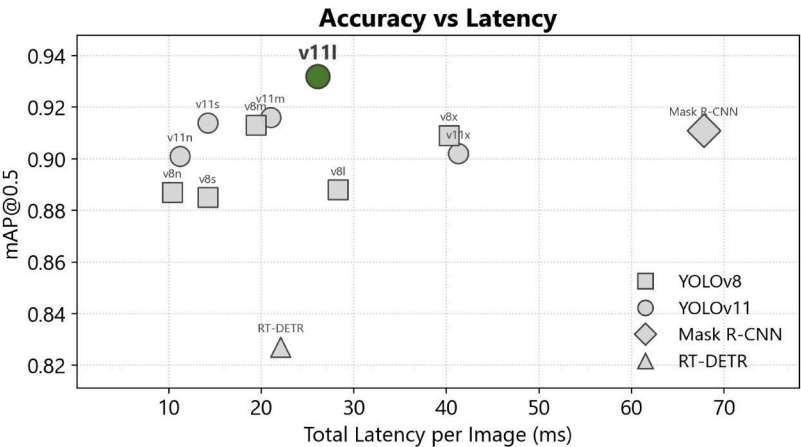


Fig. 11. Accuracy-latency trade-off for the evaluated models, using mAP@0.5 and total latency per image; RT-DETR is plotted using its box-level mAP@0.5.

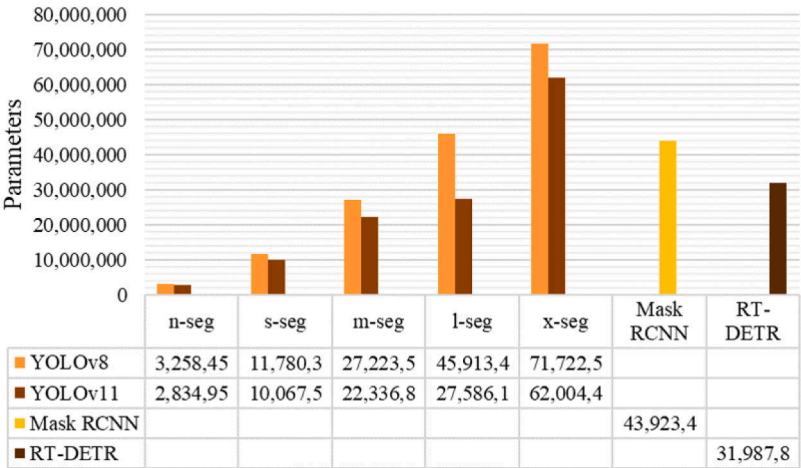


Fig. 12. Number of parameters for YOLOv8, YOLOv11, and Mask R-CNN segmentation models, and the RT-DETR.

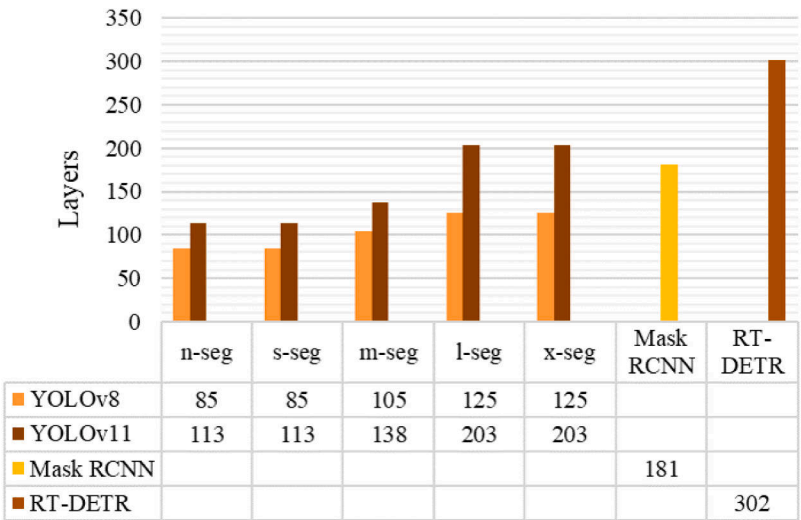


Fig. 13. Number of layers in YOLOv8, YOLOv11, and Mask R-CNN segmentation models, and the RT-DETR.

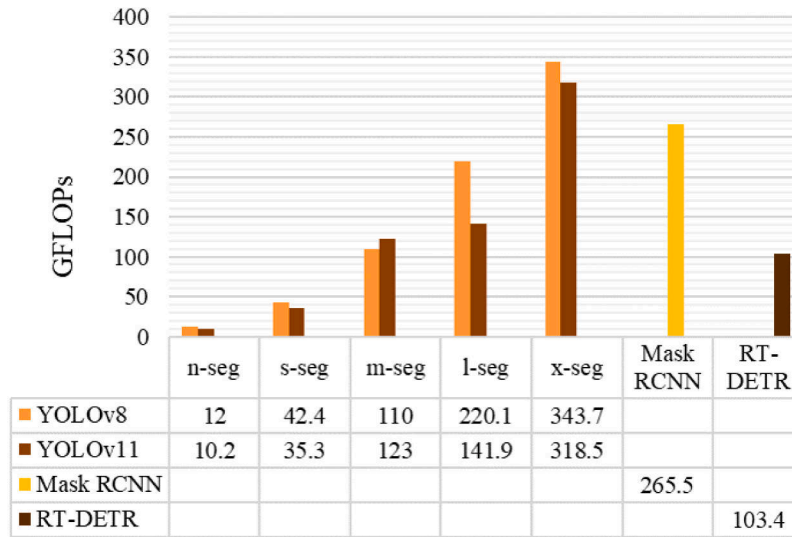


Fig. 14. Comparison of GFLOPs for YOLOv8, YOLOv11, and Mask R-CNN segmentation models, and the RT-DETR.

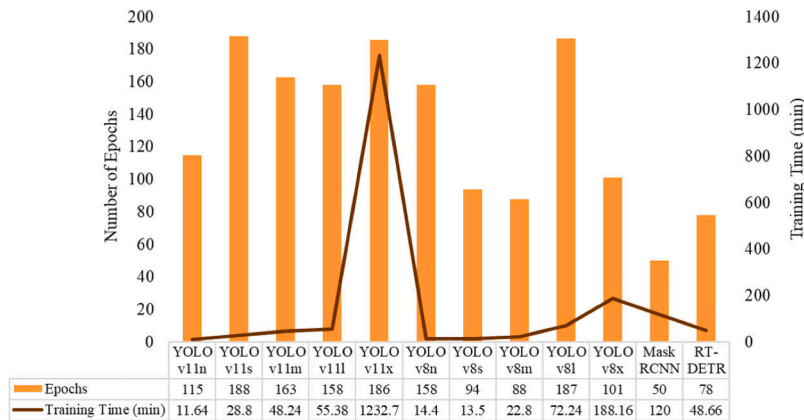


Fig. 15. Training time in minutes and number of epochs for YOLOv8, YOLOv11, and Mask R-CNN models, and the RT-DETR.

their YOLOv11 counterparts (188 and 163 epochs), suggesting that the deeper YOLOv11 architectures at these mid-scales require more parameter updates before stabilizing. However, this trend reversed at other scales: YOLOv11n converged in 115 epochs (vs. 158 for YOLOv8n), and YOLOv11l converged in 158 epochs (vs. 187 for YOLOv8l). Since YOLOv11l also ranks among the top tier of evaluated segmenters, this represents a highly favorable outcome in terms of training efficiency.

Mask R-CNN halted early after only 50 epochs, yet still accumulated a total wall-clock time of 120 minutes. Consequently, its per-epoch cost was the highest among the evaluated models (≈ 2.4 min/epoch) compared to ≈ 0.35 min/epoch for YOLOv11l (55.38 min over 158 epochs). A notable outlier is YOLOv11x, which required 1,232.7 min (~ 20.5 h) approximately $6.5\times$ the time required by YOLOv8x (188.16 min). This is likely a memory-bound artifact: YOLOv11x operated near the GPU VRAM limit, severely throttling the processing speed of each epoch. Because its mAP@0.5 remained below that of YOLOv8x, this extensive additional computational cost is not justified. Finally, the bounding-box detection, RT-DETR, converged in 78 epochs with a wall-clock time of 48.66 minutes. This demonstrates a training cost comparable to the mid-scale YOLO configurations, though without yielding the mask-level segmentation provided by the other models.

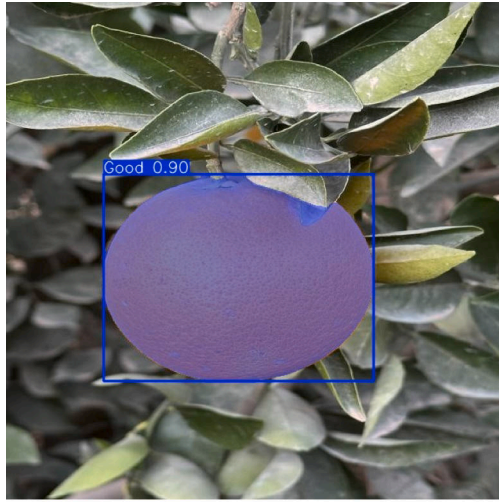
4.5. Static validation

Qualitative inference results of YOLOv11l across representative scenarios are shown in Fig. 16.

Masks closely follow fruit contours. Good fruits are correctly classified with high confidence despite leaf shadows or dust, and physical damage (rot, skin disease) is reliably assigned to the Defective class. Even touching Good/Defective pairs receive independent, non-overlapping instance boundaries, and fallen oranges on soil are detected accurately despite the strongly different background context.

4.6. Yield estimation accuracy

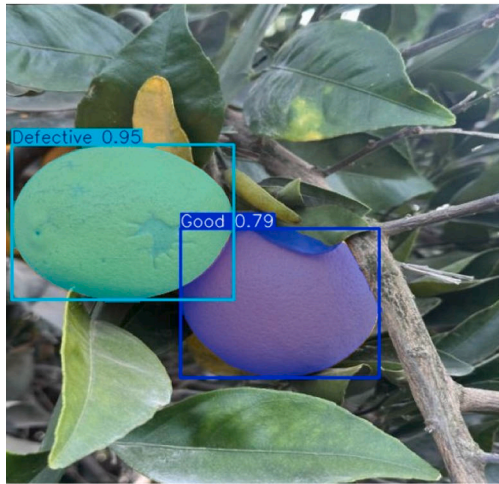
Dynamic yield estimation was evaluated using the same BoT-SORT configuration (Table 6) for every segmentation model, so that all differences in counts stem from detector stability rather than tracking. The comparison across detection paradigms is methodologically fair at the counting stage: the virtual-line mechanism of Section 3.3.2 operates exclusively on the bounding-box centroid predicted by each model (Eq. (3)), and instance masks are not utilized at any point in the tracking or counting stage. The mask-level segmenters therefore hold no advantage over the box-level RT-DETR in this comparison, as every model's



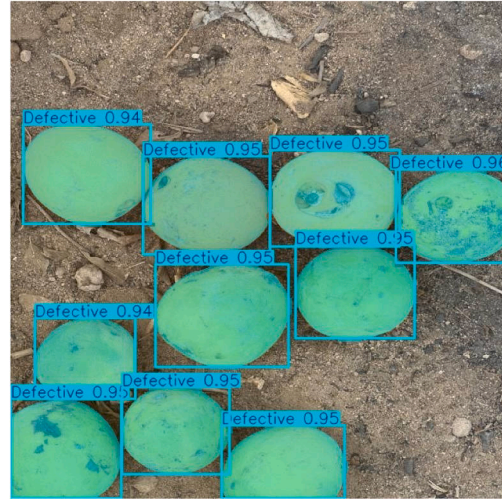
(a) Correct classification of a Good orange (Conf: 0.90).



(b) Precise identification of a Defective orange (Conf: 0.96).



(c) Instance separation of adjacent Good/Defective fruits on a tree.



(d) Multi-target detection of fallen fruit on a complex soil background.

Fig. 16. Visualized inference results of the YOLOv11 model. The blue/green overlays represent the generated segmentation masks, with bounding boxes indicating class confidence. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

predictions are processed for counting through an identical geometric representation.

4.6.1. Counting results and error analysis

From 15 videos, a subset of 6 representative videos, specifically chosen to encompass varying levels of canopy density and severe occlusion, was established as the formal evaluation benchmark. The remaining 9 videos featured highly sparse, unoccluded fruit distributions and were utilized solely for initial tracker calibration and debugging; they were excluded from the final benchmark to avoid artificially inflating the aggregate performance metrics. The BoT-SORT configuration reported in Table 6 was selected using only these nine auxiliary videos, which thus served as a development set for the tracking stage; the configuration was fixed before any of the six benchmark videos was processed, and the two video sets are fully disjoint. These auxiliary sequences include controlled, non-orchard scenes used to verify the counting logic in isolation; Fig. 18 is obtained from this set. The six benchmark videos, captured in

Egyptian orchards, span low density (20–70 oranges), medium density (100–200), and high density with heavy occlusion (400). The manual ground truth for every benchmark video was established in-field by a professional crop estimator through physical counting of the fruit in the orchard. For each benchmark video, the estimator selected a section of the orchard and defined a fixed trajectory that was traversed jointly by the estimator and the camera operator during recording. Counting was performed from the camera's vantage along this trajectory, so the reference count for each video corresponds to the fruit visible along the filmed path rather than to the total fruit load of the trees in that section; fruit fully occluded by foliage or borne on the far side of the canopy is excluded from the reference count by construction, exactly as it is absent from the model's input. The quality class of each counted fruit was assigned at the moment of counting by the same estimator, applying the same visual criteria established for the image annotation protocol (Section 3.1.2): visible disease symptoms, physical damage, and clearly abnormal discoloration or deformation. These criteria are in

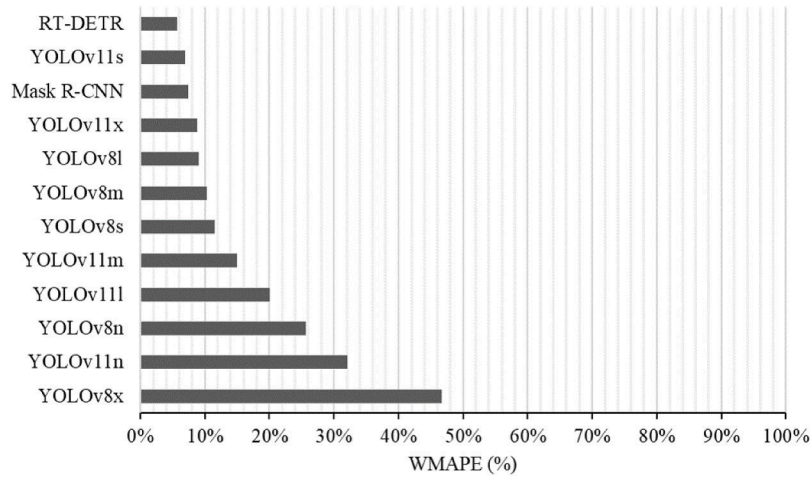


Fig. 17. Comparison of evaluated models based on Weighted Mean Absolute Percentage Error (WMAPE) for dynamic yield estimation.

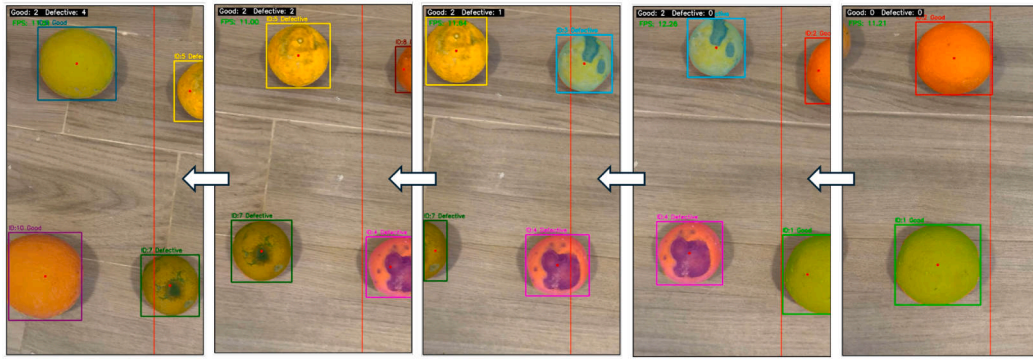


Fig. 18. Tracking success scenario in a simple scene outside the orchard; frames progress from right to left as indicated by the arrows. Each fruit retains a persistent track ID, crosses the virtual counting line exactly once, and correctly increments the Good/Defective counters.

routine commercial use at the orchard and are shared by the growers and the estimator, so the field classification and the image annotations rest on a single common definition of the Defective class. Table 8 reports Good/Defective/Total predictions against manual ground truth, Table 9 gives the signed per-video relative error and the aggregate WMAPE, and Fig. 17 ranks models by WMAPE.

Regarding the per-class composition of the dynamic counts, the low absolute Defective totals in Table 8 reflect the composition of the test footage rather than a grading failure. All validation videos were deliberately captured during the harvest season, the period in which yield estimation is operationally useful for planning, when orchards at commercial maturity naturally carry very few defective fruits (14 of 990 ground-truth fruits, a prevalence of $\approx 1.4\%$). The low prevalence of defects in the dynamic evaluation must be acknowledged as a limitation. Specifically, the dynamic grading fidelity reported in this study rests on a tiny absolute defective count; thus, the quality grading in the dynamic yield estimation pipeline requires subsequent validation on explicitly defect-rich footage.

In the target deployment, the pipeline deliberately defaults toward the Good class at long camera distances where defect cues fall below the resolvable scale. The harvested yield subsequently passes through a conveyor-belt sorting stage in which every fruit is inspected at close range; a defective fruit provisionally counted as Good in the orchard is removed downstream, whereas a spurious in-orchard Defective count would distort harvest-planning quantities. Per-class counting fidelity in the dynamic setting should therefore be interpreted as a primary screening signal complemented by downstream sorting.

The RT-DETR baseline attained the lowest overall counting error across all models, with a WMAPE of 5.66% and per-video deviations

never exceeding 15% (e.g., 199 counted against a ground truth of 200 in Video 4). This aggregate figure, however, quantifies total-count accuracy only. At the class level, RT-DETR did not resolve fruit quality reliably in the dynamic setting: it reported zero Defective fruits in Videos 1–5 (ground truth: 1, 1, 2, 4, and 3, respectively) and, conversely, assigned 20 of its 23 counted fruits to the Defective class in Video 6 against a ground truth of only 3, inverting the class composition of that scene. This behavior is consistent with the detector's comparatively low box-level Precision (0.753) in the static evaluation. The RT-DETR results should therefore be read strictly as a cross-paradigm total-yield counting demonstration.

Across the mask-level segmentation models, YOLOv11s attained the best counting stability with a WMAPE of 6.97%; for example, in Video 2 (GT = 200) it predicted 221 and in the denser Video 5 (GT = 400) it predicted 408, a 2% deviation. YOLOv8x stood at the opposite end with a WMAPE of 46.67%, returning 636 oranges in Video 5 against a ground truth of 400. Notably, YOLOv11l, among the top tier of static segmenters, consistently under-counted (WMAPE = 20.10%; 336 in Video 5), while Mask R-CNN achieved the second-best across segmentation models with WMAPE of 7.47%.

4.6.2. Influence of detection stability on tracking continuity

Because the tracker was held constant, these differences are attributable to detection stability across frames. Overcounting arises from flickering detections: in canopy clutter or under lighting changes, a model may detect a fruit, miss it in the next frame, and re-detect it afterwards; the tracker treats the reappearance as a new identity, so each flicker adds a separate crossing of the counting line, the dominant failure mode for YOLOv8x and YOLOv11n. Undercounting, seen in YOLOv11l,

Table 8

Detailed yield estimation results (Good, Defective, Total) across all six test videos. The table aggregates the Good (G), Defective (D), and Total (T) counts into a single view. The manual ground truth is provided in the first row for direct comparison.

Video	Video 1			Video 2			Video 3			Video 4			Video 5			Video 6		
	G	D	T	G	D	T	G	D	T	G	D	T	G	D	T	G	D	T
Manual	69	1	70	199	1	200	98	2	100	196	4	200	397	3	400	17	3	20
YOLOv11n	109	0	109	255	0	255	129	0	129	248	0	248	546	0	546	16	3	19
YOLOv11s	62	0	62	221	0	221	91	4	95	165	10	175	402	6	408	16	2	18
YOLOv11m	70	2	72	160	0	160	84	5	89	150	6	156	350	0	350	16	3	19
YOLOv11l	60	0	60	155	1	156	81	1	82	137	1	138	336	0	336	17	2	19
YOLOv11x	75	1	76	232	0	232	92	4	96	160	5	165	399	4	403	11	2	13
YOLOv8n	52	0	52	135	0	135	91	0	91	145	0	145	297	0	297	13	3	16
YOLOv8s	61	0	61	174	0	174	86	0	86	164	0	164	372	0	372	17	2	19
YOLOv8m	81	1	82	184	0	184	96	0	96	150	0	150	386	0	386	8	6	14
YOLOv8l	73	0	73	230	0	230	96	0	96	181	0	181	431	0	431	10	7	17
YOLOv8x	133	0	133	283	0	283	122	0	122	254	0	254	636	0	636	13	3	16
Mask R-CNN	52	0	52	191	0	191	107	0	107	179	1	180	382	1	383	15	2	17
RT-DETR	74	0	74	181	0	181	89	0	89	199	0	199	418	0	418	3	20	23

Table 9

Estimation accuracy analysis. Values represent the Relative Error (%) for each video, with the Weighted Mean Absolute Percentage Error (WMAPE) quantifying overall stability.

Model	Video 1	Video 2	Video 3	Video 4	Video 5	Video 6	WMAPE (%)
(Ground Truth)	(70)	(200)	(100)	(200)	(400)	(20)	(Lower is Better)
YOLOv11n	+55.7%	+27.5%	+29.0%	+24.0%	+36.5%	-5.0%	32.12
YOLOv11s	-11.4%	+10.5%	-5.0%	-12.5%	+2.0%	-10.0%	6.97
YOLOv11m	+2.9%	-20.0%	-11.0%	-22.0%	-12.5%	-5.0%	14.95
YOLOv11l	-14.3%	-22.0%	-18.0%	-31.0%	-16.0%	-5.0%	20.10
YOLOv11x	+8.6%	+16.0%	-4.0%	-17.5%	+0.8%	-35.0%	8.79
YOLOv8n	-25.7%	-32.5%	-9.0%	-27.5%	-25.8%	-20.0%	25.66
YOLOv8s	-12.9%	-13.0%	-14.0%	-18.0%	-7.0%	-5.0%	11.52
YOLOv8m	+17.1%	-8.0%	-4.0%	-25.0%	-3.5%	-30.0%	10.30
YOLOv8l	+4.3%	+15.0%	-4.0%	-9.5%	+7.8%	-15.0%	9.09
YOLOv8x	+90.0%	+41.5%	+22.0%	+27.0%	+59.0%	-20.0%	46.67
Mask R-CNN	-25.7%	-4.5%	+7.0%	-10.0%	-4.2%	-15.0%	7.47
RT-DETR	+5.7%	-9.5%	-11.0%	-0.5%	+4.5%	+15.0%	5.66

is the opposite pathology: the detector drops a track when the fruit becomes partially hidden and never recovers it before the counting line, so the fruit is missed entirely. Either way, a model that is highly accurate on still frames can still be unreliable on video.

YOLOv11s's low WMAPE (6.97%) indicates more temporally stable detections, letting the tracker maintain persistent identities and count each fruit exactly once. Mask R-CNN's 7.47% WMAPE stems from a different mechanism: its RPN produces a comprehensive set of candidates per frame and is therefore less prone to dropped detections during viewpoint changes. However, YOLOv11s is a much lighter single-stage model (10.1 M parameters vs. 43.9 M; 14.2 ms vs. 67.8 ms per frame, i.e., ~70 FPS vs. ~14.7 FPS), offering a combined advantage of accuracy, stability, real-time throughput, and suitability for deployment on resource-constrained edge devices that Mask R-CNN cannot match. Overall, these results establish that reliable in-field yield estimation requires temporally consistent detections, not just high per-frame mAP.

To complement these aggregate statistics with frame-level qualitative evidence, Figs. 18–20 visualize representative success and failure scenarios of the tracking-and-counting pipeline across consecutive frames, showing the assigned track IDs, the virtual counting line, and the live Good/Defective counters. Fig. 18 depicts a simple, sparsely populated scene outside the orchard canopy: every fruit retains a single persistent identity across the sequence, crosses the virtual line exactly once, and increments the corresponding Good or Defective counter, demonstrating correct identity preservation and count-once behavior. Fig. 19 illustrates the first characteristic failure mode, corresponding to the un-

dercounting behavior: the orange assigned ID 77 (dashed circle) is initially detected, then becomes totally occluded by foliage and never re-emerges before the counting line, so its track is terminated and the fruit is never counted. Fig. 20 shows the second failure mode: the bounding box of the orange assigned ID 40 (dashed circle) flickers in the frames immediately preceding the counting line and reappears afterwards, so the strict left-to-right crossing event of Eq. (3) is never registered and the fruit is likewise excluded from the count. These sequences make explicit the mechanisms through which per-frame detection instability propagates into counting error.

4.7. The proposed framework in the context of recent studies

Table 1 situates the proposed framework within recent deep learning studies on fruit detection, segmentation, and yield estimation. As the comparison shows, prior works have typically addressed these capabilities in isolation: detection-oriented citrus studies [16,18–20] operate strictly at the bounding-box level; segmentation-based studies [6,26,27] are evaluated on static single-image scenarios; and video-based yield estimation works [17,21,23,34] generally count a single generic fruit class without an accompanying quality assessment. To explicitly benchmark across architectural paradigms, this study evaluated both CNN-based segmenters and the modern, transformer-based RT-DETR detection baseline. It is also noteworthy that the dataset constructed in this study (618 original images, expanded to 1,545 augmented training images) is comparable to, and in several cases larger than, the datasets employed in these peer-reviewed studies, despite being collected entirely



Fig. 19. Tracking failure scenario 1 (track loss under total occlusion). The orange assigned ID 77 (dashed circle) is detected and subsequently disappears completely behind leaves before reaching the virtual counting line; its track is terminated and the fruit is never counted, producing an undercount.

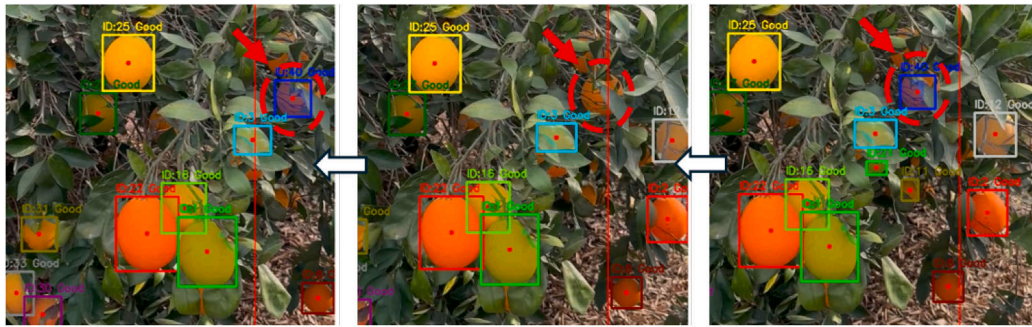


Fig. 20. Tracking failure scenario 2 (detection flicker at the counting line). The bounding box of the orange assigned ID 40 (dashed circle) is detected, flickers (transient missed detection) just before crossing the virtual line, and reappears after it; the crossing event is never registered, so the fruit is not counted.

from scratch in Egyptian orchards rather than assembled from existing public sources.

The framework developed in this study unifies these three capabilities: instance segmentation, quality-resolved classification, and tracking-based dynamic counting, within a single pipeline validated on a localized dataset from Egyptian orange orchards. The experimental results demonstrate that each stage operates at a level suitable for practical deployment. For static quality grading (Good vs. Defective), YOLOv11 achieved a mAP@0.5 of 0.932 at an inference latency of 26.1 ms per image, confirming that pixel-level grading is feasible in real time under dense occlusion and variable illumination. For dynamic yield estimation, the integration of instance segmentation with BoT-SORT tracking, camera-motion compensation, and virtual line-crossing logic enabled YOLOv11s to achieve a low counting error of 6.97% WMAPE at 14.2 ms per frame on handheld orchard videos while resolving each counted fruit into its quality class. Additionally, the bounding-box RT-

DETR baseline demonstrated strong total-count stability (WMAPE = 5.66%), although without reliable per-class resolution in the dynamic setting (Section 4.6). Taken together, these results establish the proposed framework as an end-to-end system that quantifies and grades the harvest in a single pass, extending the methods of prior studies into an integrated solution tailored to the operational conditions of the Egyptian citrus sector.

5. Conclusion

This study delivered a validated, end-to-end computer-vision framework for real-time segmentation, quality grading, and yield estimation in Egyptian orange farms, and constructed a localized dataset of 618 original images (1,545 augmented training images) and 15 videos benchmarking single-stage (YOLOv8, YOLOv11) and two-stage (Mask R-CNN) models, together with a transformer-based bounding-box de-

tection baseline (RT-DETR), under realistic occlusion, clustering, and illumination variability.

For static segmentation and quality grading, YOLOv11l was among the top tier of evaluated segmenters (mAP@0.5 = 0.932) with real-time feasibility (26.1 ms per image), and comparing favorably with Mask R-CNN, which achieved higher Recall (0.953) but at the cost of much higher latency (67.8 ms) and lower Precision. The RT-DETR's box-level mAP@0.5 (0.827) was below the mask-level mAP@0.5 of every segmentation model, though the two criteria are not directly comparable. For dynamic yield estimation with BoT-SORT and virtual line-crossing, the best model was not the most accurate one per frame: RT-DETR, despite its lowest box-level static accuracy, produced the lowest overall total-count error (WMAPE = 5.66%), although its per-class dynamic grading was unreliable; YOLOv11s achieved the lowest error among the mask-level segmentation models (WMAPE = 6.97%) while preserving quality resolution, and YOLOv11l consistently under-counted. This indicates that temporal detection stability, rather than static mAP, is the dominant factor in in-field counting reliability in this evaluation.

The principal finding of this evaluation is therefore a three-way dissociation: the highest-mAP static segmenter (YOLOv11l), the most accurate dynamic total counter (RT-DETR), and the best quality-resolved dynamic grader (YOLOv11s) are three different systems. Counting accuracy and grading capability are consequently separable properties and require separate evaluation when selecting a model for deployment. Collectively, the framework offers a practical path to modernize Egyptian citrus production by reducing reliance on subjective manual inspection and supporting harvest logistics and labor planning.

All results derive from single training runs without dispersion estimates; the counting ranking among the leading models, where RT-DETR leads YOLOv11s by 1.31 WMAPE points, therefore rests on single runs.

Future work will prioritize dataset expansion. In particular, the dynamic validation set will be extended with videos acquired in orchard blocks of higher defect prevalence: only $\approx 1.4\%$ in the present six videos were Defective, which is insufficient to characterize class-wise counting reliability. Future work will also integrate depth sensing for 3D fruit sizing and volumetric yield forecasting, and combine the fine-tuned models with mobile manipulators for autonomous inspection and robotic harvesting.

CRedit authorship contribution statement

Saleh A. Farhoud: Writing – original draft, Software, Methodology, Formal analysis; **Yasmine A. Zaghloul:** Writing – review & editing, Supervision; **Omar M. Shehata:** Writing – review & editing, Supervision.

Ethics declaration

The author(s) declare(s) that the study does not involve humans nor animal subjects.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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